

Aim: Word Count program using pig.

1.Creating a text file.

OUTPUT:

nano filename.txt:

```
[cloudera@quickstart ~]$ nano SYCS55.txt
My ordinary life
Tombstone
I am Atomic
Full drive
Magic!, fireball!
```

2.Putting the text file in a directory.

OUTPUT:

hdfs dfs -put filename.txt directory_name or path:

```
[cloudera@quickstart ~]$ hdfs dfs -put SYCS55.txt /user/cloudera/atomic
[cloudera@quickstart ~]$ hdfs dfs -ls atomic
Found 2 items
-rw-r--r-- 1 cloudera cloudera 69 2025-12-11 00:35 atomic/SYCS55.txt
-rw-r--r-- 1 cloudera cloudera 36 2025-02-01 05:55 atomic/data.txt
```

3.Writing a Pig script and creating the output directory for it.

OUTPUT:

nano load_file.pig:

```
[cloudera@quickstart ~]$ nano load_syics55.pig
data = LOAD '/user/cloudera/atomic/SYCS55.txt' as (line:chararray);
words = FOREACH data GENERATE FLATTEN(TOKENIZE(line)) as word;
grouped_words = GROUP words By word;
word count = FOREACH grouped_words GENERATE group as word, COUNT(words) as count;
STORE word_count into '/user/cloudera/output/wordcount_output5';
```

4.Running the script with the help of MapReduce.

OUTPUT:

pig -x mapreduce load_file.pig:

```
[cloudera@quickstart ~]$ pig -x mapreduce load_syics55.pig
```

```
HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features  GROUP_BY
2.6.0-cdh5.12.0 0.12.0-cdh5.12.0 cloudera 2025-12-11 00:39:59 2025-12-11 00:40:29

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime
Job_1765439353266_0002  1  1  3  3  3  3  3  3  3
user/cloudera/output/wordcount_output5,
user/cloudera/output/wordcount_output5,

Input(s):
Successfully read 6 records (451 bytes) from: "/user/cloudera/atomic/SYCS55.txt"
Output(s):
Successfully stored 12 records (92 bytes) in: "/user/cloudera/output/wordcount_output5"

Counters:
Total records written : 12
Total bytes written : 92
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
Job_1765439353266_0002

2025-12-11 00:40:29,446 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
```

5.Checking the output file in the directory.

OUTPUT:

```
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/output/wordcount_output5
Found 2 items
-rw-r--r-- 1 cloudera cloudera 0 2025-12-11 00:40 /user/cloudera/output/wordcount_output5/_SUCCESS
-rw-r--r-- 1 cloudera cloudera 92 2025-12-11 00:40 /user/cloudera/output/wordcount_output5/part-r-00000
```

6.Checking the output generated by the MapReduce.

OUTPUT:

```
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/output/wordcount_output5/part-r-00000
I 1
My 1
am 1
Full 1
life 1
drive 1
Atomic 1
Magic! 1
ordinary 1
Tombstone 1
fireball! 1
```